Knowns:

- Center Distance (C) = 5 in (Chosen)
- Pulley Diameters (D) = $D_1 = D_2$ = 3in
- Rotation Speed (n) = 1500 RPM (Chosen)
  $= 1500\left(\frac{2\pi}{60}\right) = 157.08$ rad/s
- Static Belt Tension (F) = 5.0 lbf
- Belt Material σ_{yield} = 471 psi = 3.25 MPa
- Shaft Material: 1060 Hardened Steel
  Tensile Strength yield (T_{yield}) = 70300 psi
- Coefficient of Friction (μ_s) = 1.0 (Assumption)
- Angle of Wrap = π (Due to 2 equal diameter pulleys)

Unknowns: · Minimum Diameter (d_b)

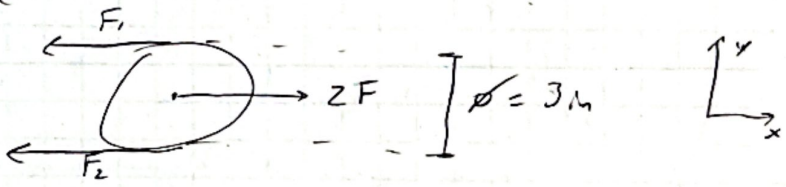
· Min Cross-Section Area of Belt (A_B)

· Forces (F_1 & F_2)

· Outer Circumference (C_{OL})

Solving For Forces: (F_1)

FBD:



$$\Sigma F_x = 2F - F_1 - F_2 = 0 \quad (eq\ 1)$$

$$F_1 = F_2$$

$$\frac{F_1}{F_2} = e^{\mu\theta} \rightarrow F_1 = F_2 e^{\mu_s \theta} \quad (eq\ 2)\ (\text{Given in Lecture Slide})$$

$$F_1 = (2F - F_1)e^{\mu_s(\pi)}$$

$$2F = F_1(1 - e^{\mu_s \theta})$$

$$F_1 = \frac{2F}{1 + e^{\mu_s(\theta)}}$$

$$F_1 = \frac{2(5\,lb_f)}{1 + e^{1(\pi)}} = \boxed{0.414\,lb_f}$$

Solving For F_2:

$$F_2 = 2F - F_1$$

$$F_2 = 2(5\,lb_f) - 0.414\,lb_f) = \boxed{9.586\,lb_f}$$

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Solving For A_B:

$$\sigma_{allow} = \frac{\sigma_{yield}}{FoS} = \frac{471\,psi}{3} = \boxed{157\,psi}$$

$$\sigma_{allow} = \frac{F_2}{A_B} \rightarrow A_B = \frac{F_2\,(FoS)}{\sigma_{allow}} = \frac{F_2}{\sigma_{allow}}$$

$$A_B = \frac{9.5858\,lb_f}{157\,psi} = \boxed{0.06105\,in^2}$$

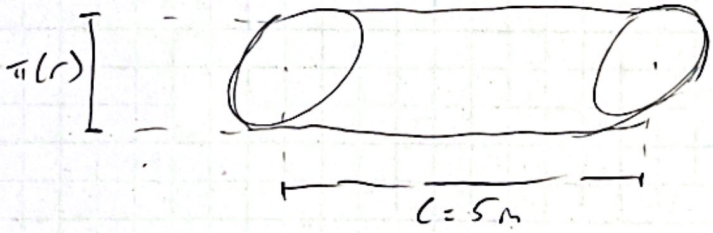
Solving For d_b:

$$A = \frac{\pi}{4} d_b^{\,2} \rightarrow d_b = \sqrt{\frac{4A_B}{\pi}}$$

$$d_B = \sqrt{\frac{4(0.06105)}{\pi}} = \boxed{0.2788\,in}$$

Solving For outer Circumference (C_{OR})

FBD:



(eq 1)

$$C_{IR} = 2C + 2(\pi r) + \pi(d_B)$$

$$= 2C + (\pi D) + \pi(d_B)$$

$$C_{IR} = 2(5in)(\pi \cdot 3in) + \pi(0.2788in)$$

$$\boxed{C_{IR} = 20.30\,in}$$

Best Selection: 5/16 in round polyurethane Rubber belt (3044K707)

Solving For Diameter of Shaft:

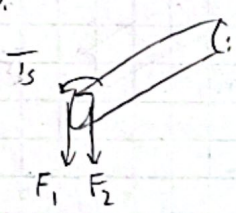
Knowns:

- Shaft Material: 1060 Hardened Steel
  $\tau_{yield} = 70300 psi$ (matweb)
- FOS = 3
- Diameter of pulley (D) = 3in
- $F_1 = 0.414\ lbf$
- $F_2 = 9.586\ lbf$

Unknowns:

- Diameter of shaft (d_s)
- Polar moment of inertia (J)

FBD:



$$\tau_{MAX} = \frac{T_s (c)}{J} = \frac{T_s \left(\frac{d_{smin}}{2}\right)}{\left(\frac{\pi d_{smin}^4}{32}\right)} = \frac{16T}{\pi d_{smin}^3}$$

(eq 1)

$$d_{smin} = \left(\frac{16 T_s}{\pi \tau_{MAX}}\right)^{1/3}$$

$$\tau_{MAX} = \frac{\tau_{yield}}{FOS}$$

$$d_{smin} = \left(\frac{16 T_s \cdot FOS}{\pi \tau_{yield}}\right)^{1/3}$$

$$T_s = \frac{D}{2}(F_2 - F_1)$$

$$d_{smin} = \left(\frac{16\left(\frac{D}{2}(F_2 - F_1)\right) \cdot FOS}{\pi \cdot \tau_{yield}}\right)^{1/3}$$

$$d_{smin} = \left[\frac{16\left(\frac{3in}{2}(1.586lb_f - 0.414lb_f) \cdot 3\right)}{\pi (70300 psi)}\right]^{1/3}$$

$$\boxed{d_{smin} = 0.1441\ in}$$

Journal Design Bearing:

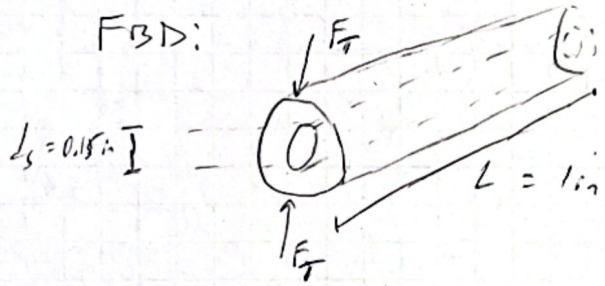
Knowns:

- $F_2 = 9.586\ lb_f$
- $F_1 = 0.414\ lb_f$
- $d_s = 0.144\ in \rightarrow 0.15\ in$ (Nominal size)
- $L = 1\ in$ (Assumption)
- Rotation Speed (n) = 1500 RPM (Chosen)

Unknowns:

- Pressure (P)
- Linear Velocity (V)
- PV

FBD:



$F_T = F_1 + F_2$ (Given in lecture slide)

Solving For Pressure:

$$P = \frac{F_1 + F_2}{d_s \cdot L} = \frac{F_T}{d_s \cdot L} = \frac{0.414\ lb_f + 9.586\ lb_f}{(0.15\ in)(1\ in)} \quad (eq\ 1)$$

$$\boxed{P = 66.7\ psi}$$

Solving For Velocity

$$V = \omega\left(\frac{d_s}{2}\right) = \left(\frac{2\pi n}{60}\right)\left(\frac{d_s \cdot 12}{2}\right) \quad \begin{matrix}(RPM \rightarrow rad/s)\\(in \rightarrow ft)\end{matrix} \quad (eq\ 2)$$

$$V = \frac{\pi (d_s)(n)}{12} = \frac{\pi (0.15\ in)(1500\ RPM)}{12}$$

$$\boxed{V = 58.905\ ft/min}$$

Solving For PV:

$$PV = P \cdot V = \left(\frac{F_1 + F_2}{d_s \cdot L}\right)\left(\frac{\pi (d_s)(n)}{12}\right) \quad (eq\ 3)$$

$$PV = \left(\frac{0.414\ lb_f + 9.586\ lb_f}{0.15\ in\ (1\ in)}\right)\left(\frac{\pi (0.15\ in)(1500\ RPM)}{12}\right)$$

$$\boxed{PV = 3928.754\ psi \cdot ft/min}$$

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Journal Bearing Material Selection:

Porous Bronze: (Material values) (Calculated values)

Bearing Pressure (P) = 4500 psi > 66.7 psi

Surface Speed (V) = 1500 ft/min > 58.905 ft/min

PV Rating = 50000 psi·ft/min > 3928.954 psi·ft/min

Frame Design

